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# Unified statistical model for "complex" and "direct" reaction mechanisms\*

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A unified statistical theory for bimolecular chemical reactions is developed. In the limit of a "direct" mechanism it becomes the usual transition state theory, which is correct for this situation, and if the reaction proceeds via a long-lived collision complex it reduces to the statistical model of Light and Nikitin. A general criterion for locating the "dividing surfaces" that are central to statistical theory is also discussed. This prescription (Keck's variational principle) is shown not only to locate the usual dividing surfaces that pass through saddle points and minima of the potential energy surface, but it also selects the critical surfaces relevant to the "orbiting" and "nonadiabatic trapping" models of complex formation.

## I. INTRODUCTION

Recent work<sup>1-5</sup> on the transition state theory<sup>6</sup> of chemical reactions has exploited the fact that the fundamental dynamical assumption of transition state theory is that the reaction mechanism is "direct." Within the framework of classical mechanics this means that transition state theory is actually *exact* if there exists a surface that divides reactant and product configuration space, through which no classical trajectory passes more than once.<sup>7</sup> Furthermore, for a simple reaction with a single saddle point separating reactants and products—such as the prototype  $H + H_2 \rightarrow H_2 + H$  reaction—this assumption of direct dynamics is indeed true for energies not too far above threshold.<sup>1,8</sup> Progress has also been made in constructing a fully quantum mechanical version of this model which, unlike the conventional quantum mechanical version of transition state theory, does not require the Hamiltonian to be separable about the saddle point.<sup>3-5</sup>

All reactions are not direct, however; many of them proceed by "complex" mechanisms, i. e., they form long-lived collision complexes.<sup>9</sup> Within the framework of classical mechanics this corresponds to classical trajectories that undergo a number of oscillations back and forth through the interaction region, and it is intuitively clear that in this situation there is no "dividing surface" through which all trajectories pass only once. The fundamental dynamical assumption of transition state theory is thus not valid.

There do exist statistical theories which are designed especially for describing complex reactions, the best known of these being associated with the work of Light<sup>10</sup> and Nikitin.<sup>11</sup> Because this approach explicitly incorporates the assumption of a long-lived collision complex, however, it becomes inapplicable when the reaction dynamics is direct.

There are thus two kinds of statistical models for bimolecular reactions—transition state theory,<sup>6</sup> which is appropriate for direct reaction dynamics, and the phase space theory<sup>10,11</sup> for complex reactions. The purpose of this paper is to construct a unified statistical theory which includes both of these models in the appropriate limiting cases.

This unified statistical theory is developed in Sec. II, and Sec. III discusses a general criterion for defining the dividing surfaces which separate reactants from products, or reactants and products from collision complexes. In addition to the simple case of a dividing surface through a saddle point or a minimum in the potential energy surface, the general criterion discussed in Sec. III is also shown to yield the dividing surfaces that are relevant to orbiting and nonadiabatic collision complexes, situations for which there are no actual saddle points on the (static) potential surface.

## II. UNIFIED STATISTICAL THEORY

For simplicity of presentation the initial development below (Secs. II, A–II, D) does not take explicit account of total angular momentum conservation; Sec. II, E gives the modifications of the final expressions that are necessary to take this into account. Also for simplicity, the figures are all drawn for the case of a collinear  $A + BC \rightarrow AB + C$  reaction for which there are only two degrees of freedom. Sections II, A and II, B first summarize the usual transition state theory<sup>6</sup> for direct reactions and the phase space theory<sup>10,11</sup> for complex reactions, respectively, and Sec. II, C then constructs their synthesis. Classical mechanics is assumed throughout the paper, although in most instances it is clear what the corresponding quantum mechanical expressions are.

### A. Direct mechanism

Figure 1 shows the sketch of a potential surface with a single saddle point separating reactants and products,

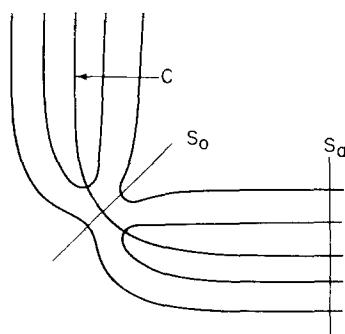


FIG. 1. Sketch of a potential energy surface for a collinear  $A + BC \rightarrow AB + C$  reaction that has a single saddle point separating reactants and products. Curve C is the "reaction path," and  $S_0$  and  $S_a$  are dividing surfaces perpendicular to it,  $S_0$  through the saddle point and  $S_a$  far out in the reactant region.

the type for which the reaction mechanism will be direct at low energy. Transition state theory gives the rate constant for this reaction as<sup>12</sup>

$$k_{b \rightarrow a}(T) = (2\pi\hbar Q_a)^{-1} \int_0^\infty dE e^{-E/kT} N_0(E), \quad (2.1)$$

where "a" and "b" denote reactants and products, respectively,  $Q_a$  is the partition function per unit volume for noninteracting reactants,  $E$  is the total energy, and  $N_0(E)$  is (proportional to) the microcanonical average of flux through the dividing surface  $S_0$  shown in Fig. 1:

$$N_0(E) = 2\pi\hbar h^{-F} \int d\mathbf{p} \int d\mathbf{q} \delta[E - H(\mathbf{p}, \mathbf{q})] \delta[f_0(\mathbf{q})] \times \frac{\partial f_0}{\partial \mathbf{q}} \cdot \mathbf{p} / \mu h \left[ \left( \frac{\partial f_0}{\partial \mathbf{q}} \right) \cdot \mathbf{p} \right]. \quad (2.2)$$

In Eq. (2.2),  $H(\mathbf{p}, \mathbf{q})$  is the total (classical) Hamiltonian function,  $f_0(\mathbf{q})$  is the function of coordinates which defines the dividing surface  $S_0$  via the equation

$$S_0: f_0(\mathbf{q}) = 0, \quad (2.3)$$

and  $h(x)$  is the unit step function,

$$h(x) = 1 \quad x > 0 \\ 0 \quad x < 0;$$

$F$  is the number of degrees of freedom, and  $(\mathbf{p}, \mathbf{q}) = (p_i, q_i)$ ,  $i = 1, \dots, F$  are the Cartesian coordinates of the system; for the collinear system illustrated in Fig. 1,  $F = 2$ . The factor  $(\partial f_0 / \partial \mathbf{q}) \cdot \mathbf{p} / \mu$  in Eq. (2.2) is the expression for  $df_0/dt$  in terms of the Cartesian coordinates and momentum  $(\mathbf{p}, \mathbf{q})$ ; if it should be more convenient to choose non-Cartesian coordinates and momenta in order to evaluate the phase space integral, then this factor is given more generally by

$$\frac{d}{dt} f_0(\mathbf{q}) = \frac{\partial f_0}{\partial \mathbf{q}} \cdot \frac{\partial H(\mathbf{p}, \mathbf{q})}{\partial \mathbf{p}}.$$

For interpretational reasons it is useful to multiply and divide the integrand of Eq. (2.1) by  $N_a(E)$ , so that it reads

$$k_{b \rightarrow a} = (2\pi\hbar Q_a)^{-1} \int_0^\infty dE e^{-E/kT} P_{b \rightarrow a}(E) N_a(E), \quad (2.4)$$

with

$$P_{b \rightarrow a}(E) \equiv N_0(E) / N_a(E), \quad (2.5)$$

where  $N_a(E)$  is the microcanonical flux through the surface  $S_a$  (cf. Fig. 1) which lies far out in the reactant region.  $N_a(E)$  is given by an expression identical to Eq. (2.2) with  $f_0(\mathbf{q})$  replaced by  $f_a(\mathbf{q})$ , the function which defines surface  $S_a$  via the equation

$$S_a: f_a(\mathbf{q}) = 0. \quad (2.6)$$

$P_{b \rightarrow a}(E)$ , defined by Eq. (2.5) as the ratio of reactive flux to incident flux, has the interpretation of an average reaction probability.<sup>13,14</sup>

It is useful to note that the flux integrals  $N_0$  and  $N_a$  introduced above have an alternate, more common interpretation than that of microcanonical flux through surfaces  $S_0$  and  $S_a$ . Thus consider the generic expression for  $N_0$  or  $N_a$ ,

$$N(E) = 2\pi\hbar h^{-F} \int d\mathbf{p} \int d\mathbf{q} \delta[E - H(\mathbf{p}, \mathbf{q})] \delta[f(\mathbf{q})] \times \frac{\partial f}{\partial \mathbf{q}} \cdot \mathbf{p} / \mu h \left( \frac{\partial f}{\partial \mathbf{q}} \cdot \mathbf{p} \right). \quad (2.7)$$

Let the coordinates  $\mathbf{q} = \{q_i\}$ ,  $i = 1, \dots, F$  be chosen so that  $q_F$  measures distance away from the surface, and  $\{q_i\}$ ,  $i = 1, \dots, F - 1$  measure displacements on the  $F - 1$  dimensional surface. The function  $f(\mathbf{q})$  is then given by

$$f(\mathbf{q}) = q_F, \quad (2.8)$$

and in properly scaled coordinates the Hamiltonian is

$$H(\mathbf{p}, \mathbf{q}) = \frac{p_F^2}{2\mu} + \sum_{i=1}^{F-1} \frac{p_i^2}{2\mu} + V(q_1, \dots, q_{F-1}, q_F), \quad (2.9)$$

so that Eq. (2.7) becomes

$$N(E) = h^{-(F-1)} \int d\mathbf{p}' \int d\mathbf{q}' \int dp_F \int dq_F \delta[E - H_{F-1}(\mathbf{p}', \mathbf{q}') - \frac{p_F^2}{2\mu}] \times \delta(q_F) \left( \frac{p_F}{\mu} \right) h(p_F), \quad (2.10)$$

where  $(\mathbf{p}', \mathbf{q}') = (p_i, q_i)$ ,  $i = 1, \dots, F - 1$  and  $H_{F-1}$  is the Hamiltonian with the  $F$ th degree of freedom removed, i. e., the Hamiltonian on the surface:

$$H_{F-1}(\mathbf{p}', \mathbf{q}') = \sum_{i=1}^{F-1} p_i^2 / 2\mu + V(q_1, \dots, q_{F-1}, 0). \quad (2.11)$$

The integrals over  $q_F$  and  $p_F$  are trivial, and one obtains

$$N(E) = h^{-(F-1)} \int d\mathbf{p}' \int d\mathbf{q}' h[E - H_{F-1}(\mathbf{p}', \mathbf{q}')], \quad (2.12)$$

which is recognized as the microcanonical partition function—i. e., the number of quantum states with energy less than or equal to  $E$ —for the reduced system of  $F - 1$  degrees of freedom. This identification of the flux integral  $N(E)$  as the number of quantum states (in the statistical mechanical sense) on the dividing surface, which is only rigorously valid within classical mechanics, is also the way in which these statistical theories are usually "quantized": Eq. (2.12) is replaced by counting discrete energy levels to determine the number of quantum states with energy below  $E$ .

## B. Complex mechanism

Figure 2 shows the sketch of a potential energy surface that has an attractive well in the A-B-C region of configuration space, corresponding to a stable triatomic molecule. In Fig. 2 this region of the collision complex is separated from reactants and products by two saddle points, but the following analysis also applies if there are no such saddle points, i. e., if the potential surface is monotonically "downhill" from reactants and products toward the well.

The "phase space theory"<sup>10,11</sup> for describing collision complexes has the form of competing first order rate processes. The rate constant is given by an expression similar to Eq. (2.1) with the flux integral  $N_0$  replaced according to

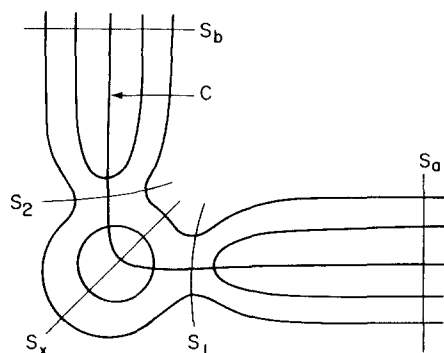


FIG. 2. Sketch of a potential energy surface for a collinear  $A + BC \rightarrow AB + C$  reaction that has an attractive well, corresponding to a stable ABC molecule, that is separated from reactants and products by saddle points. Curve C is the reaction path, and the dividing surfaces  $S_1$  and  $S_2$  cross it through the saddle points,  $S_x$  through the minimum of the potential well, and  $S_a$  and  $S_b$  in the reactant and product regions, respectively.

$$N_0(E) = \frac{N_1(E)N_2(E)}{N_1(E) + N_2(E)}, \quad (2.13)$$

where  $N_1$  and  $N_2$  are the flux integrals through surfaces  $S_1$  and  $S_2$  (cf. Fig. 2), respectively;

$$k_{b \rightarrow a} = (2\pi\hbar Q_a)^{-1} \int_0^\infty dE e^{-E/kT} \frac{N_1(E)N_2(E)}{N_1(E) + N_2(E)}. \quad (2.14)$$

[If there are no saddle point regions separating the well in the potential surface from the asymptotic regions, then the surfaces  $S_1$  and  $S_2$  are shifted to their asymptotic regions, i. e.,  $S_1 \rightarrow S_a$ ,  $S_2 \rightarrow S_b$ , so that in Eq. (2.14)  $N_1 \rightarrow N_a$ ,  $N_2 \rightarrow N_b$ .] It is customary in the statistical model of long-lived collision complexes to think of  $N_1$  and  $N_2$  as the number of internal states on surfaces  $S_1$  and  $S_2$ , but as noted in the last paragraph of Sec. II. A, this is equivalent to their interpretation as fluxes.

Again, for interpretational reasons, one multiplies and divides the integrand in Eq. (2.14) by common factors so that the rate constant in Eq. (2.4) is given by Eq. (2.4), but where the net reaction probability is now given by

$$P_{b \rightarrow a}(E) = \frac{P_{b \rightarrow x} P_{x \rightarrow a}}{P_{b \rightarrow x} + P_{a \rightarrow x}}, \quad (2.15)$$

where

$$P_{x \rightarrow a}(E) = N_1(E)/N_a(E), \quad (2.16a)$$

$$P_{a \rightarrow x}(E) = N_1(E)/N_x(E), \quad (2.16b)$$

$$P_{b \rightarrow x}(E) = N_2(E)/N_x(E), \quad (2.16c)$$

$N_1$ ,  $N_2$ , and  $N_x$  being the flux integrals through surfaces  $S_1$ ,  $S_2$ , and  $S_x$ , respectively, in Fig. 2.  $P_{x \rightarrow a}$ ,  $P_{a \rightarrow x}$ , and  $P_{b \rightarrow x}$  have the interpretations of the probability of reactants  $a$  forming a collision complex  $x$ , the probability that the complex decays back to reactants, and the probability that the complex decays to products  $b$ , respectively. (Note that the flux integral  $N_x$  actually cancels out in the equation for  $P_{b \rightarrow a}$ .) The net reaction probability  $P_{b \rightarrow a}$  thus has the usual form<sup>15</sup> of the probability that reactants from a collision complex,  $P_{x \rightarrow a}$ , times the branching ratio  $P_{b \rightarrow x}/(P_{a \rightarrow x} + P_{b \rightarrow x})$  for the complex breaking up to give products.

### C. Unified theory

To summarize the results of Secs. II. A and II. B, the rate constant is in general given by

$$k_{b \rightarrow a} = (2\pi\hbar Q_a)^{-1} \int_0^\infty dE e^{-E/kT} P_{b \rightarrow a}(E) N_a(E), \quad (2.4)$$

where in the direct case one has

$$P_{b \rightarrow a}^{\text{direct}} = (N_0/N_a), \quad (2.5)$$

and in the complex case

$$P_{b \rightarrow a}^{\text{complex}} = \frac{P_{b \rightarrow x} P_{x \rightarrow a}}{P_{b \rightarrow x} + P_{a \rightarrow x}} = \frac{(N_2/N_x)(N_1/N_a)}{(N_2/N_x) + (N_1/N_x)}, \quad (2.15')$$

where the various flux integrals  $N(E)$  are all of the form of Eq. (2.7) for the various surfaces indicated in Figs. 1 and 2.

$P_{b \rightarrow a}^{\text{direct}}$  in Eq. (2.5) is clearly inapplicable to the complex situation, and  $P_{b \rightarrow a}^{\text{complex}}$  in Eq. (2.15) is also incorrect in the direct limit. To see this, suppose the potential surface in Fig. 2 is distorted so that the well is raised and the saddle point through which  $S_2$  passes is diminished, so much so that the well and saddle point merge and disappear. In this limit the potential surface reverts to one with a single saddle point, i. e., the direct type in Fig. 1; surfaces  $S_2$  and  $S_x$  in Fig. 2 merge so that  $N_2 \equiv N_x$ , and  $N_1$ , the flux through the remaining saddle point, is what is designated  $N_0$  in Fig. 1. The reaction probability given by the complex prescription in Eq. (2.15') thus becomes

$$P_{b \rightarrow a}^{\text{complex}} \rightarrow \frac{(1)(N_0/N_a)}{1 + (N_0/N_a)}, \quad (2.17)$$

which is not the correct result, Eq. (2.5), for the direct case.

The reason that Eq. (2.15') fails in the direct limit is clear; the existence of a long-lived collision complex necessarily implies that

$$P_{b \rightarrow x}, P_{a \rightarrow x} \ll 1, \quad (2.18)$$

otherwise the collision complex would not be long lived. In the direct limit considered above, however,  $P_{b \rightarrow x} \rightarrow 1$ , which is physically correct but which violates Eq. (2.18), an assumption implicit in Eq. (2.15).

To construct a theory capable of describing both direct and complex limits, therefore, it is necessary to define a net reaction probability  $P_{b \rightarrow a}$  that is valid for all values of  $P_{x \rightarrow a}$ ,  $P_{a \rightarrow x}$ , and  $P_{b \rightarrow x}$  between 0 and 1. This is accomplished by using the probability branching analysis of Hirschfelder and Wigner.<sup>16</sup> A sketch of the branching picture is given in Fig. 3, and the net reaction probability is the sum of the reaction probabilities for all the possible branches:

$$P_{b \rightarrow a} = \sum_{n=0}^{\infty} P_{b \rightarrow x} (1 - P_{b \rightarrow x})^n (1 - P_{a \rightarrow x})^n P_{x \rightarrow a}. \quad (2.19)$$

The  $n$ th term in Eq. (2.19) corresponds to a collision complex that has survived  $n$  oscillations back and forth between surfaces  $S_1$  and  $S_2$ ; the probability of this

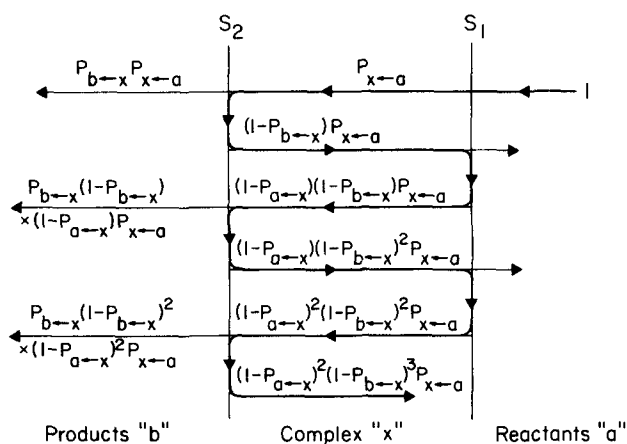


FIG. 3. Sketch of the flow of probability from reactants  $a$  through the complex  $x$  to products  $b$ .  $P_{x-a}$ ,  $P_{a-x}$ , and  $P_{b-x}$  are the transition probabilities between the indicated regions for a given encounter at one of the dividing surfaces,  $S_1$  and  $S_2$ .

branch is the product of the probability of forming the collision complex from reactants,  $P_{x-a}$ , the probability that the complex does not decay to reactants or products on  $n$  encounters at surfaces  $S_1$  and  $S_2$ ,  $(1 - P_{b-x})^n \times (1 - P_{a-x})^n$ , and the probability that the complex decays to products,  $P_{b-x}$ . This geometric series is easily summed to give<sup>16</sup>

$$P_{b-a} = \frac{P_{b-x}P_{x-a}}{1 - (1 - P_{b-x})(1 - P_{a-x})} = \frac{P_{b-x}P_{x-a}}{P_{b-x} + P_{a-x} - P_{b-x}P_{a-x}} \quad (2.20a)$$

$$= \frac{(N_2/N_x)(N_1/N_a)}{(N_2/N_x) + (N_1/N_x) - (N_2/N_x)(N_1/N_x)}. \quad (2.20b)$$

The Appendix shows that Eq. (2.20) also results from a WKB treatment of transmission through a double barrier potential.

Equation (2.20) is seen to differ from Eq. (2.15) only by the term in the denominator which is quadratic in probability. For a long-lived collision complex, for which Eq. (2.18) holds, this quadratic term will be negligible compared to the two linear terms, so that Eq. (2.15) is recovered in the complex limit. On the other hand, in the direct limit described above—for which  $N_2 \rightarrow N_x$  so that  $P_{b-x} \rightarrow 1$ , and  $N_1 \equiv N_0$ —Eq. (2.20) gives

$$P_{b-a} = \frac{(1)(N_0/N_a)}{1 + P_{a-x} - (1)P_{a-x}} = (N_0/N_a),$$

which is Eq. (2.5), the correct result for the case of a direct reaction.

Equation (2.20), therefore, is the desired result; it reduces to the correct direct and complex expressions in the appropriate limits. Inserting Eq. (2.20) into Eq. (2.4) gives the following expression for the rate constant in this unified statistical theory:

$$k_{b-a} = (2\pi\hbar Q_a)^{-1} \times \int_0^\infty dE e^{-E/kT} \frac{N_2(E)N_1(E)}{N_2(E) + N_1(E) - N_2(E)N_1(E)/N_x(E)}, \quad (2.21)$$

and one sees that, unlike Eq. (2.14), the flux integral  $N_x(E)$ —the flux “out of” the collision complex—does not cancel out in this more general expression.

#### D. Collinear harmonic case

In many actual applications of these statistical expressions the flux integrals  $N(E)$  are evaluated by approximating the potential surface as quadratic about the saddle points (and minima). Thus it is interesting and illustrative to see the explicit form taken by the reaction probability in the simplest case, a collinear  $A + BC \rightarrow AB + C$  reaction within the harmonic approximation.

In this case the generic flux integral  $N(E)$  is

$$N(E) = (E - V)/\hbar\omega, \quad (2.22)$$

where  $V$  is the potential energy of the saddle point (or minimum), and  $\omega$  is the harmonic frequency for the direction *along* the relevant surface  $S$ , i. e., for the vibrational motion perpendicular to the reaction coordinate.

For the case of a single barrier, as in Fig. 1, transition state theory, Eq. (2.5), thus gives

$$P_{b-a}^{\text{direct}}(E) = \frac{\omega_a}{\omega_0} \frac{E - V_0}{E}, \quad (2.23)$$

where  $\omega_a$  and  $\omega_0$  are the vibrational frequencies of the reactant molecule  $BC$  and the symmetric stretch of the “activated complex,” respectively,  $V_0$  is the potential energy of the saddle point, and where the zero of total energy is chosen as the bottom of the reactant valley. Equation (2.23) is a well-known result.<sup>13,14</sup> [If it happens that  $\omega_a > \omega_0$ , as is usually the case, the reader will notice that Eq. (2.23) gives  $P_{b-a} > 1$  as  $E \rightarrow \infty$ ; this is an example of the fact that simple transition state theory usually fails at high energy.]

For the potential surface sketched in Fig. 2, which has a stable collision complex separated from reactants and products by saddle points, the phase space model for a long-lived collision complex, Eq. (2.15'), gives

$$P_{b-a}^{\text{complex}} = \frac{\omega_a}{\omega_1 + \omega_2} \frac{(E - V_1)(E - V_2)}{E(E - \bar{V})} \quad (2.24a)$$

with

$$\bar{V} \equiv \frac{\omega_2 V_1 + \omega_1 V_2}{\omega_2 + \omega_1}, \quad (2.24b)$$

where  $V_1$  and  $\omega_1$ , and  $V_2$  and  $\omega_2$ , are the potential and frequency for saddle points 1 and 2, respectively. If the potential surface is symmetric—i. e.,  $V_2 = V_1$ ,  $\omega_2 = \omega_1$ —the Eq. (2.24) simplifies the

$$P_{b-a}^{\text{complex}} = \frac{1}{2} \frac{\omega_a}{\omega_1} \frac{E - V_1}{E}, \quad (2.25)$$

which is similar to Eq. (2.23) but with the additional factor of  $\frac{1}{2}$ ; this is the “transmission coefficient” implicit in the phase space theory<sup>10,11</sup> since it assumes that the complex breaks up equally likely to give products or reactants.

The unified statistical theory, Eq. (2.20), on the other hand, gives the reaction probability in the symmetrical case as

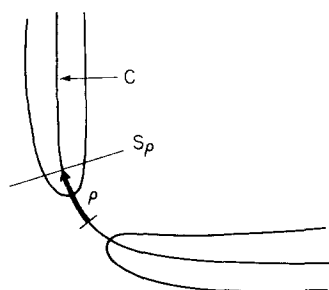


FIG. 4. Sketch of a potential energy surface indicating the reaction path  $C$  and reaction coordinate  $\rho$ , the arc length along  $C$ .  $S_p$  is a dividing surface that crosses  $C$  at the value  $\rho$  of the reaction coordinate.

$$P_{b \rightarrow a}^{\text{unified}} = \left( 2 - \frac{\omega_x}{\omega_1} \frac{E - V_1}{E - V_x} \right)^{-1} \frac{\omega_a}{\omega_1} \frac{E - V_1}{E}, \quad (2.26)$$

where  $V_x$  and  $\omega_x$  are the potential energy and frequency for the minimum of the potential (cf. Fig. 2). This expression may be thought of as Eq. (2.25) with the addition of an energy-dependent "transmission coefficient"

$$\left( 2 - \frac{\omega_x}{\omega_1} \frac{E - V_1}{E - V_x} \right)^{-1}. \quad (2.27)$$

For low energies and strongly bound complexes one has

$$(E - V_1)/(E - V_x) \ll 1,$$

so that

$$2 - \frac{\omega_x}{\omega_1} \frac{E - V_1}{E - V_x} \approx 2,$$

and Eq. (2.26) thus reduces to Eq. (2.25) in this limit. In the direct limit that the well in the potential surface becomes shallow and disappears—i. e.,  $V_x \rightarrow V_1$ ,  $\omega_x \rightarrow \omega_1$ —one has

$$2 - \frac{\omega_x}{\omega_1} \frac{E - V_1}{E - V_x} \rightarrow 2 - 1 = 1,$$

so that Eq. (2.26) also reduces to the correct result, Eq. (2.23), in this limit.

### E. Reactions in three dimensions; conservation of total angular momentum

To take explicit account of total angular momentum conservation, which is necessary when treating reactions in three dimensional space, Eq. (2.4) is replaced by

$$k_{b \rightarrow a} = (2\pi\hbar Q_a)^{-1} \int_0^\infty dE e^{-E/kT} \int_0^\infty dJ 2J P_{b \rightarrow a}(E, J) N_a(E, J), \quad (2.28)$$

where  $P_{b \rightarrow a}(E, J)$  is the net reaction probability for total energy  $E$  and total angular momentum  $J$ .  $P_{b \rightarrow a}(E, J)$  is given in terms of flux integrals by precisely the same expressions as in Secs. II, A–II, C but where the flux integrals are now for a specific value of total angular momentum as well as total energy and have the generic form

$$N(E, J) = 2\pi\hbar h^{-F} \int dp \int dq \delta[E - H(p, q)] \delta[J - \mathcal{J}(p, q)] \times \delta[M - \mathfrak{M}(p, q)] \delta[f(q)] \frac{\partial f}{\partial q} \cdot p / \mu \hbar \left( \frac{\partial f}{\partial q} \cdot p \right). \quad (2.29)$$

$\mathcal{J}(p, q)$  and  $\mathfrak{M}(p, q)$  in Eq. (2.29) are the total angular momentum and its component along a space-fixed axis, respectively, expressed in terms of the phase space integration variables  $(p, q)$ ;  $M$  may have any value between  $-J$  and  $J$  since the flux integrals are independent of it.

### III. SPECIFICATION OF DIVIDING SURFACES

A central role is played in the expressions of the previous section by the dividing surfaces through which the various flux integrals are calculated. This section explores in more detail the precise way in which these surfaces can be defined. It is important that the general definition applies not only to the simple case that the potential energy surface has actual saddle points and minima, but also includes the dividing surfaces which define collision complexes within the orbiting and non-adiabatic trapping models, which are unrelated to saddle points or minima of the (static) potential energy surface.

The most general and rigorous way of defining the dividing surfaces is by Keck's<sup>17</sup> variational principle, i. e., so that  $N_0(E)$  in Fig. 1, and  $N_1$  and  $N_2$  in Fig. 2, are relative *minima* of flux in the reactive direction, i. e., so that they are the "bottlenecks" of the reaction. It is then also natural to define  $S_x$  in Fig. 2 so that  $N_x$  is a relative *maximum* of flux.

To see this more clearly, consider the continuous family of dividing surfaces  $S_p$ , sketched in Fig. 4, which are parameterized by the "reaction coordinate"  $\rho$ , the arc length along the reaction path (Curve  $C$  in Fig. 4).  $S_p$  is the dividing surface perpendicular to Curve  $C$  at the distance  $\rho$  along  $C$ .

Considering first the collinear  $A + BC \rightarrow AB + C$  reaction, suppose the potential energy surface is harmonic in the degree of freedom perpendicular to the reaction path; in natural collision coordinates<sup>18</sup> the potential surface is given within this harmonic approximation by

$$V(\rho, u) = V_0(\rho) + \frac{1}{2}k(\rho)u^2, \quad (3.1)$$

where  $u$  is the vibrational coordinate, the degree of freedom perpendicular to the reaction path, and  $k(\rho)$  is the force constant for this vibrational mode at position  $\rho$  along the reaction path. As discussed in Sec. II, D, the flux integral through surface  $S_p$ ,  $N_p(E)$ , is then given by

$$N_p(E) = \frac{E - V_0(\rho)}{\hbar\omega(\rho)}, \quad (3.2)$$

where  $\omega(\rho)$  is the vibrational frequency for the  $u$  motion at position  $\rho$

$$\omega(\rho) = \sqrt{\frac{k(\rho)}{\mu}}. \quad (3.3)$$

If it is first assumed that  $\omega(\rho)$  does not change significantly with  $\rho$ , then it is clear from Eq. (3.2) that  $N_p$  has a relative minimum (maximum) of those values of  $\rho$  for which  $V_0(\rho)$  has a relative maximum (minimum). For the potential surfaces sketched in Figs. 1 and 2,  $N_p(E)$  has the form sketched in Figs. 5a and 5b, respectively. The flux integrals  $N_0$  of Fig. 1, and  $N_1$ ,

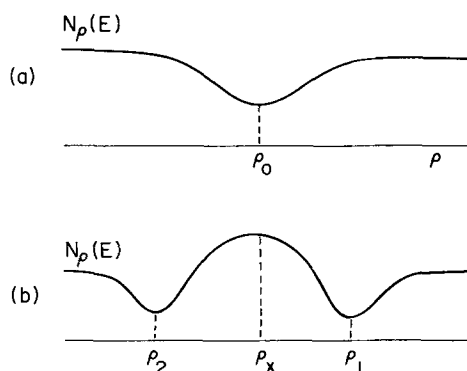


FIG. 5. The flux integral  $N_\rho(E)$  through surface  $S_\rho$ , as a function of  $\rho$ , for a potential surface of the type in (a) Fig. 1 and (b) Fig. 2.

$N_2$ , and  $N_x$  of Fig. 2, are the values of  $N_\rho$  at  $\rho_0$  [in Fig. 5(a)] and  $\rho_1$ ,  $\rho_2$ ,  $\rho_x$  [in Fig. 5(b)]. As suggested above, therefore,  $N_0$ ,  $N_1$ , and  $N_2$  are relative minima (bottle-necks) of the flux along the reaction path, and  $N_x$  is a relative maximum.

Equation (3.2) can also be used to illustrate the phenomenon of nonadiabatic trapping<sup>19,20</sup> which results if  $\omega(\rho)$  varies strongly along the reaction path. Consider a potential surface of the type sketched in Fig. 1. Typically the vibrational mode "relaxes" in the region of the saddle point, i. e.,  $\omega(\rho)$  has a minimum at  $\rho = 0$ . [For the  $H + H_2$  reaction<sup>20</sup>  $\omega(0) \approx \frac{1}{2}\omega(\infty)$ .] Figure 6 shows a sketch of the two factors that contribute to  $N_\rho(E)$ ,  $E - V_0(\rho)$  and  $1/\omega(\rho)$ , and one can clearly see that if  $1/\omega(\rho)$  is sufficiently sharply peaked, then  $N_\rho(E)$  will have the same qualitative shape as in Fig. 5(b), i. e., there can be two bottleneck regions (minima in  $N_\rho$ ) on either side of the actual saddle point of the potential surface. This qualitative explanation of nonadiabatic trapping in terms of relaxation of the vibrational mode about the saddle point is usually<sup>20,21</sup> based on the assumption of vibrational adiabaticity. It is very interesting to see here that it also comes out of a purely sta-

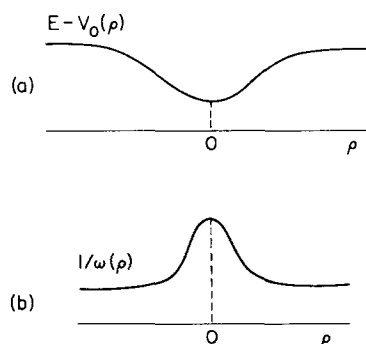


FIG. 6. Sketch of the two factors that contribute to Eq. (3.2), for a potential surface of the type in Fig. 1.

tistical theory when the variational criterion is used to locate the relevant dividing surfaces.

Finally, to show that the variational criterion also predicts the correct dividing surface for the orbiting model of collision complexes it is necessary to consider a three dimensional system,  $A + BC \rightarrow AB + C$ , for example. As discussed in Sec. II, E, for reactions in three dimensions the flux integrals must be defined for a given value of total angular momentum  $J$  as well as total energy  $E$ . In the orbiting model the reaction coordinate is taken to be the radial coordinate  $R$ , the distance between  $A$  and the center of mass of  $BC$ , and the interaction potential between  $A + BC$  is assumed to be a function only of  $R$  (usually the van der Waals type  $-C/R^6$ ). The  $A + BC$  system in its overall center of mass has six degrees of freedom, and the 12 dimensional integral in Eq. (2.29) is most conveniently evaluated if the six pair of coordinates and momenta are chosen as the following action-angle variables<sup>22,23</sup>:

- $(P, R)$ : radial motion of  $A - BC$ ;
- $(l, q_l)$ : orbital angular momentum of  $A$  relative to  $BC$ ;
- $(j, q_j)$ : rotational angular momentum of  $BC$ ;
- $(n, q_n)$ : vibrational motion of  $BC$ ;
- $(J, q_J)$ : total angular momentum;
- $(M, q_M)$ : projection of total angular momentum.

The flux integral [Eq. (2.29)] through the dividing surface located at the value  $R$  of the radial coordinate is then given by (setting  $\hbar = 1$ )

$$N_R(E, J) = (2\pi)^{-5} \int_0^\infty dn \int_0^\infty dj \int_0^\infty dJ' \int_{-J'}^{J'} dM' \int_{J'-j}^{J'+j} dl \int_{-\infty}^\infty dP \int_0^\infty dR' \int_0^{2\pi} dq_n \int_0^{2\pi} dq_j \int_0^{2\pi} dq_J \int_0^{2\pi} dq_M \int_0^{2\pi} dq_l \times \delta(E - H) \delta(J - J') \delta(M - M') \delta(R - R') \frac{h}{\mu} h(P), \quad (3.4)$$

where the Hamiltonian in these variables is<sup>23</sup>

$$H = \frac{P^2}{2\mu} + \frac{l^2}{2\mu R^2} + V(R) + \epsilon(n, j), \quad (3.5)$$

$\epsilon(n, j)$  being the vibration/rotation energy of  $BC$  in terms of the "quantum numbers" (actually action variables)  $n$  and  $j$ . Since the integrand of Eq. (3.4) is independent of the five angle variables, the integral over them is trivial, and the delta functions allow one to do the integrals over  $R$ ,  $P$ ,  $J'$ , and  $M'$ , yielding

$$N_R(E, J) = \int_0^\infty dn \int_0^\infty dj \int_{|J-j|}^{J+j} dl h \left( E - V(R) - \epsilon(n, j) - \frac{l^2}{2\mu R^2} \right); \quad (3.6)$$

the limits on the  $l$  integral are the usual "triangle relation" involving the angular momenta  $l$ ,  $j$ , and  $J$ .

In most applications of the orbiting model it is also assumed that most of the angular momentum is orbital, i. e.,

$$l \gg j, \quad (3.7a)$$

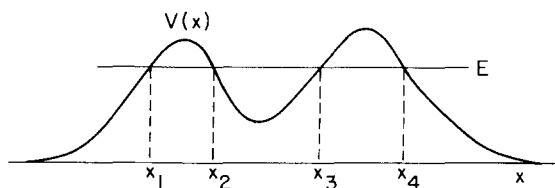


FIG. 7. A one dimensional double barrier potential function;  $E$  is the collision energy.

so that

$$l \approx J, \quad (3.7b)$$

whereby Eq. (3.6) becomes

$$N_R(E, J) = \int_0^\infty dn \int_0^\infty dj \, 2j \, h \left[ E - V(R) - \frac{J^2}{2\mu R^2} - \epsilon(n, j) \right] \quad (3.8a)$$

$$\equiv W \left( E - V(R) - \frac{J^2}{2\mu R^2} \right), \quad (3.8b)$$

where the function  $W(\bar{E})$  is defined by

$$W(\bar{E}) = \int_0^\infty dn \int_0^\infty dj \, 2j \, h[\bar{E} - \epsilon(n, j)]. \quad (3.9)$$

$W(\bar{E})$  is thus the number of quantum states of the diatomic molecule BC with energy below  $\bar{E}$ , i. e., the microcanonical partition function of BC.  $W(\bar{E})$  is a strictly increasing function of  $\bar{E}$ —if BC is approximated as a rigid rotor–harmonic oscillator, for example,  $W(\bar{E}) \propto \bar{E}^2$ —so that the value of  $R$  which makes  $N_R(E, J)$  an extremum is

$$0 = \frac{\partial}{\partial R} N_R(E, J) \\ = -W' \left( E - V(R) - \frac{J^2}{2\mu R^2} \right) \frac{\partial}{\partial R} \left( V(R) + \frac{J^2}{2\mu R^2} \right), \quad (3.10)$$

or since  $W' > 0$ , the condition is

$$0 = \frac{\partial}{\partial R} \left( V(R) + \frac{J^2}{2\mu R^2} \right); \quad (3.11)$$

i. e., the dividing surface which makes the flux a minimum is at that value of  $R$  for which the “effective potential”  $V(R) + J^2/2\mu R^2$  has a maximum, and this is the usual criterion of the orbiting model. It should be noted, though, that even when the approximations in Eq. (3.7) are not valid, Eq. (3.6) is still the correct expression for  $N_R(E, J)$  (although the integrals are somewhat messier to evaluate)<sup>15</sup> and the optimum choice for  $R$  can still be determined variationally, i. e., by Eq. (3.10).

One thus sees that the variational criterion for choosing the dividing surface provides a unified prescription that correctly describes these several quite different physical situations.

#### IV. SUMMARY AND CONCLUDING REMARKS

Section II has presented a unified statistical theory of bimolecular reactions that includes as special limiting cases the usual transition state theory, which is appropriate for a direct reaction mechanism, and the phase space theory of Light and Nikitin, which is ap-

propriate for reactions that proceed via a long-lived collision complex. The theory is completely nonempirical, i. e., it involves the potential energy surface of the reacting system but has no empirical parameters. Since this unified theory is correct in the limit of a completely direct mechanism and that of a long-lived collision complex, one can perhaps reasonably hope that it will also be useful for intermediate situations.

The variational criterion<sup>17</sup> for choosing the relevant dividing surfaces which appear in the theory is discussed in Sec. III. In addition to yielding the usual dividing surfaces that are related to saddle points and minima in the potential energy surface, this general criterion is seen also to produce the critical surfaces relevant to the orbiting model and also that which describes nonadiabatic trapping. It thus appears that this unified statistical theory, together with the variational criterion for locating the dividing surfaces, is capable of describing a wide variety of different kinds of dynamical phenomena in chemical reactions.

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#### APPENDIX: DERIVATION OF EQ. (2.20) VIA THE WKB APPROXIMATION

It is interesting to see that the fundamental equation of the unified statistical model, Eq. (2.20), is also obtained by a WKB treatment of transmission by a double barrier potential.

Consider the one-dimensional potential sketched in Fig. 7, where  $E$  is the collision energy.  $\theta_1$  and  $\theta_2$  are the barrier penetration integrals through barriers 1 and 2, and  $\phi$  is the phase integral across the potential well:

$$\theta_1 = \int_{x_1}^{x_2} dx \sqrt{2\mu[V(x) - E]}/\hbar^2, \\ \theta_2 = \int_{x_3}^{x_4} dx \sqrt{2\mu[V(x) - E]}/\hbar^2, \\ \phi = \int_{x_2}^{x_3} dx \sqrt{2\mu[E - V(x)]}/\hbar^2,$$

The WKB approximation for the transmission probability has been considered by Child,<sup>24</sup> and the general result can be written in the following simple form:

$$P(E) = \frac{p_1 p_2}{1 + (1 - p_1)(1 - p_2) + 2\sqrt{1 - p_1}\sqrt{1 - p_2}\cos 2\phi}, \quad (A1)$$

where  $p_1$  and  $p_2$  are the WKB tunneling probabilities through barriers 1 and 2:

$$p_i = 1/(1 + e^{2\theta_i}), \quad i = 1, 2. \quad (A2)$$

As a function of energy  $E$ ,  $P(E)$  in Eq. (A1) has a resonance structure that is uninteresting for purposes of determining an *average* transmission probability.

This average transmission probability  $\bar{P}(E)$ —which corresponds to averaging  $P(E)$  over an energy interval that contains many individual resonances but which is still small compared to the energy variation of  $\bar{P}(E)$ —can be obtained by averaging Eq. (A1) over  $\phi$ :

$$\bar{P}(E) = \frac{1}{\pi} \int_0^\pi d\phi \frac{p_1 p_2}{1 + (1 - p_1)(1 - p_2) + 2\sqrt{1 - p_1}\sqrt{1 - p_2} \cos 2\phi}, \quad (\text{A3})$$

and it is easy to show that this gives

$$\bar{P}(E) = \frac{p_1 p_2}{1 - (1 - p_1)(1 - p_2)} = \frac{p_1 p_2}{p_1 + p_2 - p_1 p_2}, \quad (\text{A4})$$

which is the form of Eq. (2.20). If  $p_1, p_2 \ll 1$ , then Eq. (A4) becomes

$$\bar{P}(E) \simeq p_1 p_2 / (p_1 + p_2),$$

corresponding to a long-lived collision complex, and if  $p_i \rightarrow 1$ , then it becomes

$$\bar{P}(E) \simeq p_j \quad (j \neq i),$$

the direct limit.

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